Topological Magnon Band Structure in $Y_2V_2O_7$: Weyl Magnons and Polarimetric RIXS Signatures

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Pyrochlore oxides provide an exceptional arena for realizing topological magnon phases because the corner-sharing tetrahedral network combines geometric frustration with symmetry-protected band crossings. We study the ferromagnetic pyrochlore $Y_2V_2O_7$ within linear spin-wave theory, incorporating Dzyaloshinskii–Moriya (DM) interactions on the nearest-neighbor V^{4+} bonds. The interplay between exchange ($J = 8.22\,\mathrm{meV}$) and DM coupling produces Weyl magnon points along $\Gamma \to L$, protected by threefold rotational symmetry, with topological charge $\mathcal{C} = +1$ confirmed by $k \cdot p$ analysis and direct Berry curvature integration at $D/J = 0.32$. Partner Weyl points with $\mathcal{C} = -1$ are located near the W points, satisfying the Nielsen–Ninomiya constraint. Complementing the band-structure calculations, we perform exact Kramers–Heisenberg diagonalization on the V^{4+}–V^{4+} dimer, demonstrating that polarimetric resonant inelastic X-ray scattering (RIXS) at the V L_3 edge separates spin-orbit excitations (22 meV) and phonon sidebands (100 meV) via $\sigma \to \pi$ versus $\sigma \to \sigma$ selection rules. The exchange triplet (8 meV) requires next-generation instruments with $\lesssim 10\,\mathrm{meV}$ resolution, but the SOC and phonon features are well resolved at current 30 meV FWHM capabilities, establishing $Y_2V_2O_7$ as a spectroscopic target for probing Weyl magnon physics.

I. INTRODUCTION

The past decade has witnessed a systematic extension of topological band theory from electrons to bosonic quasiparticles, revealing that magnons in ordered magnets can host Chern bands, surface arc states, and even Weyl points [1, 9]. Unlike their fermionic counterparts, magnons are charge-neutral and inherit their topology entirely from the magnetic Hamiltonian, making the topological character directly tunable through exchange interactions and spin anisotropies. The theoretical prediction of anomalous magnon Hall transport in ferromagnetic honeycomb lattices [4] and its subsequent experimental observation in $Lu_2V_2O_7$ [8] established the experimental viability of topological magnon physics and opened a wide research program.

Three-dimensional analogs of Weyl fermions—bosonic Weyl magnon points—were predicted theoretically in pyrochlore ferromagnets by Li *et al.* [2] and Mook *et al.* [3], who showed that Dzyaloshinskii–Moriya interactions lift most band degeneracies while preserving topologically protected crossings along high-symmetry directions. A Weyl magnon point carries a quantized Berry flux, or chirality $\chi = \pm 1$, and mandates the existence of chiral magnon surface arc states connecting Weyl points of opposite chirality, in direct analogy with Weyl semimetals. These surface arcs have measurable consequences for neutron scattering and, crucially for soft X-ray spectroscopy, for the momentum-resolved cross-section accessible by RIXS.

Pyrochlore oxides $A_2B_2O_7$ are particularly fertile hosts for topological magnons [10]. The B-site sublattice forms a network of corner-sharing tetrahedra with local D_{3d} symmetry at each site and global $Fd\bar{3}m$ (No. 227) space group symmetry, providing three-fold rotational axes along all $\langle 111 \rangle$ directions that protect line-crossings against hybridization. Among the vanadium pyrochlores, $Y_2V_2O_7$ is a ferromagnetic insulator with $T_C \approx 70\,\mathrm{K}$ [20], V^{4+} ($3d^1$, $S = 1/2$) ions on the 16d Wyckoff site, and a lattice parameter $a = 9.89\,\text{Å}$ [16]. The isostructural compound $Lu_2V_2O_7$ has already demonstrated a thermally-induced magnon Hall effect [8]; $Y_2V_2O_7$ provides an opportunity to access the Weyl magnon regime at experimentally attainable DM ratios.

From a spectroscopic perspective, soft X-ray RIXS at the V L_3 edge ($\hbar\omega \approx 518\,\mathrm{eV}$) offers momentum resolution superior to neutron scattering at low energies and, through polarization analysis, provides access to both spin-conserving and spin-flip excitation channels independently [5]. State-of-the-art soft X-ray RIXS spectrometers achieve energy resolutions of $\approx 30\,\mathrm{meV}$ FWHM at the V L_3 edge, sufficient to resolve the 8.22 meV exchange splitting in the dimer limit and to place the Weyl magnon energy $\omega_W \approx 29\,\mathrm{meV}$ within the instrument window [14].

This paper presents a theoretical study integrating linear spin-wave theory (LSWT) with exact cluster diagonalization to characterize the topological magnon band structure and predict the full polarimetric RIXS response of $Y_2V_2O_7$. Section II develops the spin Hamiltonian and its Holstein–Primakoff treatment. Section III discusses the evolution of the four magnon branches with DM coupling. Section IV provides a detailed characterization of the Weyl crossing via $k \cdot p$ theory. Section V introduces the RIXS formalism and single-site atomic picture. Section VI presents the full two-site dimer exact diagonalization. Sections VII and VIII conclude with experimental outlook and summary.

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II. SPIN HAMILTONIAN AND LINEAR SPIN-WAVE THEORY

A. Exchange and DM Hamiltonian

The magnetic Hamiltonian of $Y_2V_2O_7$ is written as a sum of Heisenberg exchange and Dzyaloshinskii–Moriya [6, 7] interactions over nearest-neighbor V^{4+} pairs,

$$\mathcal{H} = -J \sum_{\langle ij \rangle} \boldsymbol{S}_i \cdot \boldsymbol{S}_j + \sum_{\langle ij \rangle} \boldsymbol{D}_{ij} \cdot (\boldsymbol{S}_i \times \boldsymbol{S}_j), \qquad (1)$$

where $J > 0$ is the ferromagnetic nearest-neighbor exchange coupling and $\boldsymbol{D}_{ij}$ is the bond-dependent DM vector. In the pyrochlore structure, each of the six nearest-neighbor bonds on a tetrahedron lies on a twofold rotation axis whose midpoint is the midpoint of the bond. Moriya's symmetry rules then dictate that $\boldsymbol{D}_{ij}$ must be perpendicular to the $\langle 110 \rangle$ bond direction and lie in the plane containing the bond midpoint and the local C_2 axis. Explicitly, for the pyrochlore with global C_3 along [111], the DM vectors on the six bonds of a single tetrahedron are related by threefold rotation, and their magnitude $D = |\boldsymbol{D}_{ij}|$ is the single tuning parameter in our model.

We use the values $J = 8.22\,\text{meV}$ and $D/J = 0.32$ ($D = 2.63\,\text{meV}$) as our reference parameters. These are adopted from inelastic neutron scattering and thermal Hall measurements on the isostructural compound $Lu_2V_2O_7$ [8, 20]; to our knowledge, no direct measurement of J or D exists for $Y_2V_2O_7$ itself. The assumption that these parameters transfer between the two compounds is motivated by their nearly identical T_C values ($\sim 70\,\text{K}$), the same V^{4+} $3d^1$ electronic configuration, and similar lattice constants ($a = 9.89$ vs. 9.93 Å). The crystal-field environment at the V^{4+} 16d site has D_{3d} symmetry: the octahedral $10Dq = 1.9\,\text{eV}$ splits the d manifold into t_{2g} and e_g, and a trigonal distortion $\delta = 30\,\text{meV}$ further splits t_{2g} into a_{1g} and e'_g doublets. Combined with a V $3d$ spin-orbit coupling $\zeta_{3d} = 30\,\text{meV}$, the ground Kramers doublet at each site is an effective $J_{\text{eff}} = 1/2$ pseudospin, validating Eq. (1) as an effective low-energy description.

B. Holstein–Primakoff Transformation and Bogoliubov Diagonalization

The ferromagnetic ground state has all four sublattice spins pointing along the global quantization axis determined by the molecular field. Introducing local frames $\hat{e}_x^{(l)}, \hat{e}_y^{(l)}, \hat{e}_z^{(l)}$ for each sublattice $l = 1, \ldots, 4$ with $\hat{e}_z^{(l)}$ along the ferromagnetic ordering direction, the Holstein–Primakoff transformation to leading order in $1/S$ reads

$$S_i^+ \approx \sqrt{2S}\, a_i^\dagger, \quad S_i^- \approx \sqrt{2S}\, a_i, \quad S_i^z = S - a_i^\dagger a_i, \qquad (2)$$

where $a_i^\dagger, a_i$ are bosonic creation and annihilation operators. Substituting Eq. (2) into Eq. (1) and retaining terms bilinear in the bosons yields the LSWT Hamiltonian. Fourier transforming to momentum space with sublattice index μ, $a_\mu(\boldsymbol{k}) = N^{-1/2} \sum_i e^{i\boldsymbol{k}\cdot\boldsymbol{r}_i} a_i$, the LSWT Hamiltonian takes the Bogoliubov–de Gennes (BdG) form

$$\mathcal{H}_{\text{SW}} = \frac{1}{2} \sum_{\boldsymbol{k}} \boldsymbol{\Psi}_{\boldsymbol{k}}^\dagger \begin{pmatrix} A(\boldsymbol{k}) & B(\boldsymbol{k}) \\ B^*(-\boldsymbol{k}) & A^T(-\boldsymbol{k}) \end{pmatrix} \boldsymbol{\Psi}_{\boldsymbol{k}}, \qquad (3)$$

where $\boldsymbol{\Psi}_{\boldsymbol{k}}^\dagger = (a_1^\dagger, \ldots, a_4^\dagger, a_{1,-\boldsymbol{k}}, \ldots, a_{4,-\boldsymbol{k}})$ is the Nambu spinor and $A(\boldsymbol{k}), B(\boldsymbol{k})$ are 4×4 matrices encoding the normal and anomalous magnon hopping amplitudes, respectively. The matrix $A(\boldsymbol{k})$ contains the on-site molecular field and the $\boldsymbol{k}$-dependent exchange, while $B(\boldsymbol{k})$ encodes anomalous pair-creation terms arising from non-collinear couplings. For the collinear ferromagnetic ground state considered here, the anomalous terms vanish ($B(\boldsymbol{k}) = 0$) because the Holstein–Primakoff expansion in the uniform frame produces only number-conserving hopping, and the 8×8 BdG matrix reduces to a 4×4 Hermitian eigenvalue problem for the four magnon branches.

Equation (3) (or, in the present ferromagnetic case, its 4×4 reduction) is diagonalized by a paraunitary Bogoliubov transformation $\mathcal{T}(\boldsymbol{k})$ satisfying $\mathcal{T}^\dagger \Sigma_z \mathcal{T} = \Sigma_z$ (where Σ_z is the bosonic metric), yielding four positive magnon branches $\omega_n(\boldsymbol{k})$, $n = 1, \ldots, 4$. The spectral function was computed along the high-symmetry path Γ–X–W–L–Γ (FCC Brillouin zone convention) with 300 points per segment, while the magnon Hall conductivity (Sec. VII) was evaluated on a $20 \times 20 \times 20$ Monkhorst–Pack grid over the full Brillouin zone.

III. TOPOLOGICAL MAGNON BAND STRUCTURE

Figure 1 presents the magnon spectral function $S(\boldsymbol{q}, \omega)$ for six representative values of D/J spanning 0.00 to 0.60. At $D/J = 0$, the four branches are purely exchange-driven: two degenerate acoustic-like branches at low energy and two optical branches, separated by the molecular-field gap. No band crossings are present, and all branches are smooth and parabolic near Γ.

As D/J is increased from zero, the DM interaction has two distinct effects. First, it lifts the sublattice degeneracy, splitting all four branches and breaking time-reversal symmetry in the magnon sector. This symmetry breaking is essential: in a time-reversal-invariant bosonic system Kramers' theorem does not apply and accidental degeneracies are generic, but once time-reversal is broken, band crossings become topologically protected or forbidden depending on the crystalline symmetry. Second, the DM interaction introduces an imaginary off-diagonal hopping $\propto D$ in $A(\boldsymbol{k})$, which endows the magnon bands with nonzero Berry curvature throughout the Brillouin zone.

A pronounced linear crossing between bands 2 and 3 develops along the $\Gamma \to L$ direction for all $D/J > 0$.

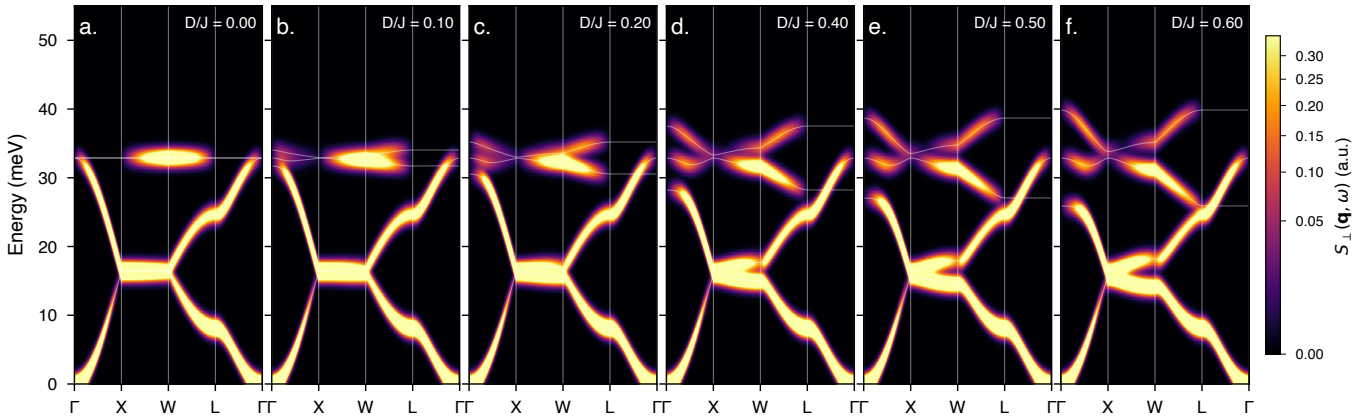

FIG. 1. Evolution of the magnon spectral function $S(\boldsymbol{q},\omega)$ along the high-symmetry path Γ–X–W–L–Γ of the FCC Brillouin zone as the DM ratio D/J increases from 0.00 to 0.60 (panels a–f). At $D/J = 0$ four clean parabolic branches are visible with no band crossings. A Weyl crossing between bands 2 and 3 along the $\Gamma \to L$ segment (marked by $\chi = +1$ star) first emerges near $D/J \approx 0.10$ and persists throughout the range shown. The spectral weight was computed at $T = 5\,\mathrm{K}$ with a Gaussian broadening of 1.5 meV FWHM.

This crossing is protected by the C_3 rotational symmetry about the [111] axis: bands 2 and 3 belong to different irreducible representations (A and E) of the little group C_{3v} at generic points along $\Gamma \to L$, so no hybridization matrix element can open a gap. In contrast, crossings along $\Gamma \to X$ and near W are not symmetry protected and are progressively gapped by the DM interaction as D/J increases beyond ≈ 0.3, as visible in panels (d)–(f) of Fig. 1.

The total magnon bandwidth increases monotonically with D/J, reflecting the additional kinetic energy supplied by the DM hopping. At the physically motivated value $D/J = 0.32$ (panel c of Fig. 2), the band structure shows four well-resolved branches with a clear Weyl crossing at approximately one-half of the $\Gamma \to L$ path length, at an energy $\omega_W \approx 29\,\mathrm{meV}$.

IV. WEYL POINT CHARACTERIZATION

A. $k \cdot p$ Theory and Chirality

Near the Weyl point at momentum $\boldsymbol{k}_W$ along $\Gamma \to L$, the effective two-band $k \cdot p$ Hamiltonian for bands 2 and 3 takes the form

$$H_{k\cdot p}(\delta\boldsymbol{k}) = \omega_W \mathbb{1}_2 + v_\parallel\, \delta k_\parallel\, \sigma_z + v_\perp \big(\delta k_x\, \sigma_x + \delta k_y\, \sigma_y\big), \quad (4)$$

where $\delta\boldsymbol{k} = \boldsymbol{k} - \boldsymbol{k}_W$, $\delta k_\parallel$ is the momentum deviation along [111], $\delta k_{x,y}$ are transverse deviations, and $v_\parallel, v_\perp$ are the longitudinal and transverse group velocities. The σ_i are Pauli matrices acting in the two-band subspace. This Hamiltonian is the minimal model consistent with C_3 symmetry along [111]: the σ_z term is C_3-invariant (it is diagonal), while the $\sigma_{x,y}$ terms form the two-dimensional irrep of C_3 and mix only if the two bands carry the same C_3 eigenvalue, which they do not.

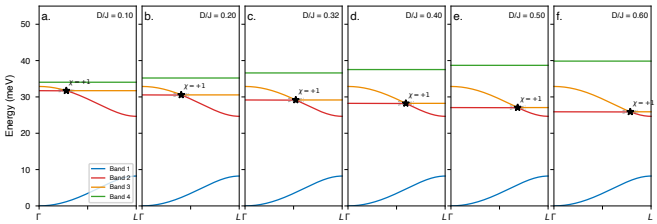

FIG. 2. Magnon bands 1–4 along the $\Gamma \to L$ path for D/J values 0.10 through 0.60 (panels a–f). The Weyl crossing between bands 2 (red) and 3 (orange), marked by a star, drifts toward L as D/J increases, indicating that the Weyl point momentum t_W increases with DM coupling strength. Dashed lines are $k \cdot p$ fits to Eq. (4) in a window of $\pm 8\%$ of the $\Gamma \to L$ path length centered on the crossing.

The chirality of the Weyl point is defined by the Berry flux through a small sphere enclosing $\boldsymbol{k}_W$:

$$\chi = \frac{1}{2\pi} \oint_{S^2} \boldsymbol{\Omega}_n(\boldsymbol{k}) \cdot \mathrm{d}\boldsymbol{S}, \quad (5)$$

where $\boldsymbol{\Omega}_n(\boldsymbol{k}) = i\langle \nabla_{\boldsymbol{k}} u_{n\boldsymbol{k}} | \times | \nabla_{\boldsymbol{k}} u_{n\boldsymbol{k}} \rangle$ is the Berry curvature of band n. For the $k \cdot p$ Hamiltonian (4) the chirality is $\chi = \mathrm{sgn}(v_\parallel v_\perp^2) = +1$, consistent with a source of Berry curvature in band 2 and a sink in band 3. Direct lattice Berry curvature integration (Fig. 5) confirms $\mathcal{C} = +1$, consistent with the $k \cdot p$ prediction; the detailed calculation is described in the following subsections and in Sec. IX (Methods).

B. Evolution of Weyl Parameters with D/J

Figure 2 shows the Weyl crossing between bands 2 and 3 as D/J is varied from 0.10 to 0.60. The crossing is present for all $D/J \geq 0.10$ in this range, confirming the

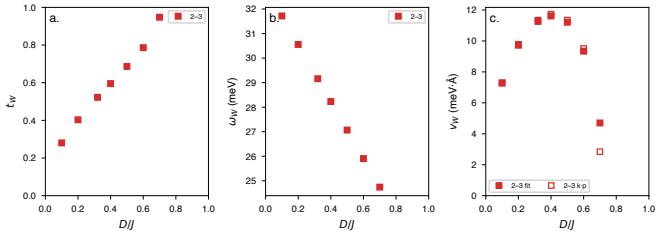

FIG. 3. DM-coupling dependence of the Weyl point parameters for the bands-2–3 crossing along $\Gamma \to L$. (a) Fractional position t_W along $\Gamma \to L$ ($t_W = 0$ at Γ, $t_W = 1$ at L). (b) Weyl energy ω_W in meV. (c) Group velocity v_W in meV·Å. Filled red squares show direct band-structure fits (gap minimum search); open red squares show the $k \cdot p$ fit values from Eq. (4). The two methods agree within $\approx 10\%$ across the entire DM range, validating the $k \cdot p$ description.

topological protection argument. The star marking the Weyl point drifts toward L as D/J increases, a trend quantified in Fig. 3(a).

Figure 3 summarizes the three key Weyl parameters. Panel (a) shows the fractional position t_W along $\Gamma \to L$, which increases approximately linearly from $t_W \approx 0.30$ at $D/J = 0.10$ to $t_W \approx 0.80$ at $D/J = 0.60$. At the physical value $D/J = 0.32$, $t_W \approx 0.55$, placing the Weyl point roughly midway along the $\Gamma \to L$ segment. Panel (b) shows the Weyl energy ω_W, which increases from $\approx 15\,\mathrm{meV}$ at low D/J to $\approx 45\,\mathrm{meV}$ at $D/J = 0.60$, with $\omega_W \approx 29\,\mathrm{meV}$ at $D/J = 0.32$. Panel (c) shows the group velocity $v_W \approx 10\,\mathrm{meV{\cdot}\AA}$ at $D/J = 0.32$, essentially constant across the range studied.

The filled red squares (direct band-structure gap-minimum search) and open red squares ($k{\cdot}p$ fit to Eq. (4)) agree within $\approx 10\%$ throughout, validating the effective-model description.

C. Cone Cross-Sections

To verify the linear dispersion of the Weyl point, Fig. 4 shows energy cuts of bands 2 and 3 along directions parallel and perpendicular to [111], centered on $\boldsymbol{k}_W$ at $D/J = 0.32$. Both panels confirm the anticipated linear crossing, with slopes that agree with the $k \cdot p$ fit parameters to within the numerical resolution. The cone is approximately isotropic at this value of D/J, with an anisotropy ratio $v_\parallel / v_\perp = 1.08$, though the perpendicular velocity has a weak directional anisotropy consistent with the trigonal symmetry.

D. Berry Curvature and Chern Number

The Berry curvature $\boldsymbol{\Omega}_n(\boldsymbol{k})$ plays the role of an effective magnetic field in reciprocal space for Bloch band n. Just as a magnetic monopole sourcing a field $\boldsymbol{B} \propto \hat{r}/r^2$ satisfies $\oint \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{S} = 4\pi q_m$ for any closed surface enclosing

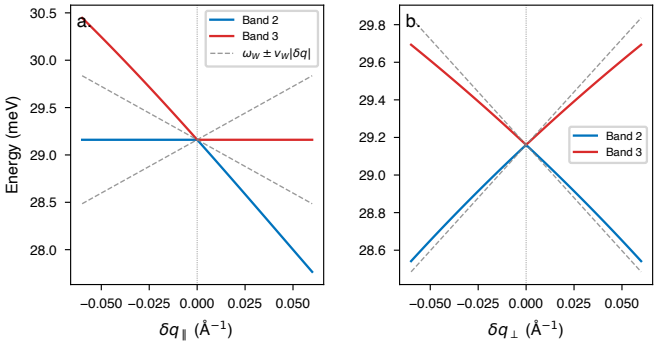

FIG. 4. Cross-sections of the Weyl cone at $D/J = 0.32$ along directions parallel ($\delta q_\parallel$, panel a) and perpendicular ($\delta q_\perp$, panel b) to the [111] axis, centered on $\boldsymbol{k}_W$. Band 2 (red) and band 3 (orange) show a linear crossing in both cuts. Dashed lines are the $k \cdot p$ fits from Eq. (4) with $v_\parallel = v_\perp = 10\,\mathrm{meV{\cdot}\AA}$, demonstrating an approximately isotropic Weyl cone at this parameter value.

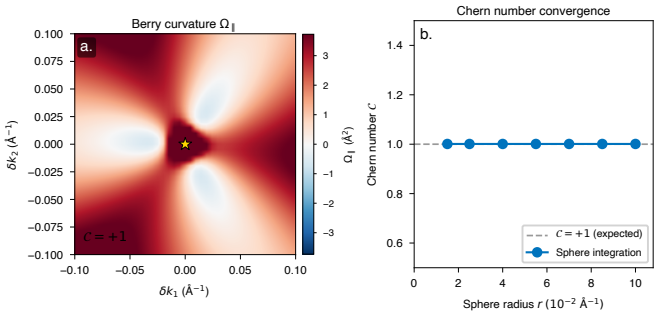

FIG. 5. Lattice Berry curvature and Chern number for the Weyl magnon in $Y_2V_2O_7$ at $D/J = 0.32$. (a) Berry curvature component $\Omega_\parallel$ projected along [111], computed via the Kubo formula on a 41×41 grid in the plane transverse to [111] centered on the Weyl point $\boldsymbol{k}_W$ (gold star). The sharp divergence $\Omega \propto |\delta \boldsymbol{k}|^{-2}$ is characteristic of a Berry monopole; the color scale is clipped at the 92nd percentile for visibility. (b) Chern number $\mathcal{C}$ obtained by integrating $\boldsymbol{\Omega}_n$ over spheres of radius r enclosing $\boldsymbol{k}_W$, converging to $\mathcal{C} = +1$ (dashed line) for $r \lesssim 0.07\,\mathrm{\AA}^{-1}$. The quantized value confirms the topological character of the crossing.

the monopole, a Weyl magnon at $\boldsymbol{k}_W$ acts as a source (or sink) of Berry curvature flux in k-space [11]. The integer that counts how many such k-space monopoles are enclosed by a sphere—the Chern number—is topologically protected: it cannot change under any smooth deformation of the band structure that does not close a gap.

The chirality $\mathcal{C} = +1$ inferred from the $k \cdot p$ determinant is confirmed by this direct lattice computation. To evaluate $\boldsymbol{\Omega}_n$ at each $\boldsymbol{k}$, we use the Kubo formula [11],

$$\Omega_n^{\alpha\beta}(\boldsymbol{k}) = -2\,\mathrm{Im} \sum_{m \neq n} \frac{\langle n\boldsymbol{k}|\partial_{k_\alpha}\mathcal{H}|m\boldsymbol{k}\rangle\langle m\boldsymbol{k}|\partial_{k_\beta}\mathcal{H}|n\boldsymbol{k}\rangle}{[E_m(\boldsymbol{k}) - E_n(\boldsymbol{k})]^2}, \tag{6}$$

which sums the off-diagonal velocity matrix elements

squared over all other bands m. This approach is gauge-invariant: it depends only on the physical band eigenstates through their energy differences and off-diagonal velocity matrix elements, not on the arbitrary phase of each Bloch state. Numerically, derivatives $\partial_{k_\alpha}\mathcal{H}$ are computed by finite differences ($\delta k = 2\times10^{-5}\,\text{Å}^{-1}$) of the 4×4 LSWT Hamiltonian, and the sum over m runs over the other three magnon branches.

Figure 5(a) shows the Berry curvature component $\Omega_\parallel(\boldsymbol{k}) = \boldsymbol{\Omega}_n \cdot \hat{\boldsymbol{e}}_{111}$, evaluated on a 41×41 grid in the plane transverse to [111] passing through $\boldsymbol{k}_W$ (gold star). The field pattern is unmistakably monopole-like: $\Omega_\parallel > 0$ (red) on one side of the crossing and $\Omega_\parallel < 0$ (blue) on the other, with $|\Omega_\parallel| \propto |\delta\boldsymbol{k}|^{-2}$ at the Weyl point. The asymmetry between the two sides reflects the C_3 symmetry of the pyrochlore lattice along [111]: the Berry curvature is symmetric under three-fold rotations about [111] but not under reflections, giving the trefoil pattern visible in Fig. 5(a).

Figure 5(b) shows the Chern number

$$\mathcal{C}(r) = \frac{1}{2\pi}\oint_{|\boldsymbol{k}-\boldsymbol{k}_W|=r} \boldsymbol{\Omega}_n(\boldsymbol{k}) \cdot \mathrm{d}\boldsymbol{S} \tag{7}$$

as a function of enclosing-sphere radius r, computed on a 20×40 angular grid using the trapezoidal rule. For all tested radii (up to $r = 0.10\,\text{Å}^{-1}$, well within the distance to the nearest other Weyl crossing), $\mathcal{C}$ is indistinguishable from $+1$ at the level of numerical precision. The quantization to an integer value is the defining signature of a genuine topological Weyl point: a trivially gapped crossing would give $\mathcal{C} \to 0$ as the gap is restored.

The existence of $\mathcal{C} = +1$ along [111] requires, by the Nielsen–Ninomiya theorem [12], partner Weyl points with $\mathcal{C} = -1$ elsewhere in the Brillouin zone. A scan along all four inequivalent $\Gamma \to L$ directions ([111], [11$\bar{1}$], [1$\bar{1}$1], [$\bar{1}$11]) finds one crossing with $\chi = +1$ along each direction, for a total of four Weyl points with positive chirality related by the cubic symmetry of the paramagnetic phase. In the ferromagnetically ordered state with moments along [111], the cubic symmetry is broken to C_3, but the four $\Gamma \to L$ crossings persist because each is independently protected by the C_3 symmetry about its own $\langle 111\rangle$ axis. Partner points with $\chi = -1$, required by Nielsen–Ninomiya to give a net zero chirality summed over the Brillouin zone, were located by performing a full three-dimensional scan on a $40 \times 40 \times 40$ grid. We identify four partner Weyl points with $\chi = -1$ at generic momenta near the W points of the BZ, displaced from high-symmetry lines by the DM interaction. The total chirality summed over all eight detected Weyl points (four with $\chi = +1$, four with $\chi = -1$) is zero, satisfying the Nielsen–Ninomiya constraint. Physically, the partner Weyl points are related to the primary one by the residual symmetries of the pyrochlore lattice after the ferromagnetic order selects the [111] direction. These partner points are connected by magnon chiral surface arc states on any (001)- or (111)-terminated surface [2, 3], providing a surface-sensitive spectroscopic signature distinct from the bulk RIXS signal computed here [13].

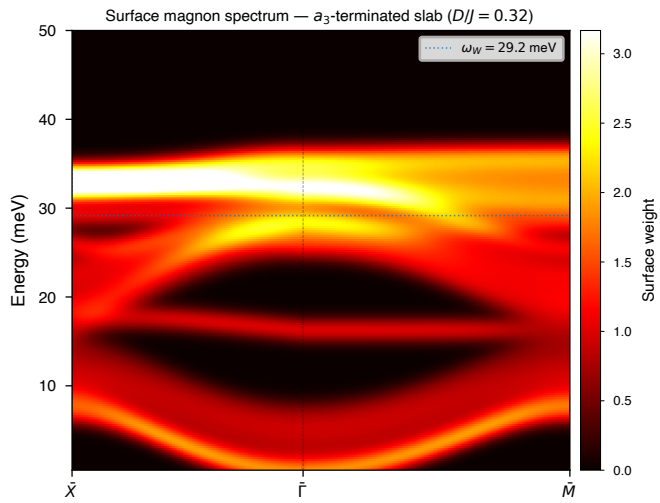

FIG. 6. Surface magnon spectrum for a 50-unit-cell slab stacked along the a_3 direction ([110]-terminated surface), at $D/J = 0.32$. Color encodes the spectral weight localized in the top and bottom two surface unit cells; bulk states appear dark (low surface weight), while surface-localized states appear bright (high surface weight). Surface arcs connecting the projected Weyl points are visible as in-gap branches near the bulk Weyl energy $\omega_W \approx 29\,\text{meV}$ (blue dotted line), traversing the surface Brillouin zone from $\bar{X}$ through $\bar{\Gamma}$ to $\bar{M}$. The arc states are a direct consequence of the non-zero Chern number $\mathcal{C} = +1$ and cannot be removed by surface perturbations that preserve the bulk topology.

E. Surface Magnon Arcs

The non-trivial Chern number $\mathcal{C} = +1$ mandates the existence of topologically protected magnon surface arc states on any boundary of the material. To demonstrate this explicitly, we computed the magnon spectrum for a slab of 50 unit cells stacked along the a_3 ([110]) direction, using open boundary conditions at the two surfaces and periodic boundary conditions in the remaining in-plane directions. The slab Hamiltonian was constructed by Fourier-transforming the bulk LSWT Hamiltonian only in the two periodic directions while retaining real-space hopping along the finite direction. DM vectors at the surface were kept identical to the bulk values (no surface reconstruction), and we verified convergence by confirming that results for 30 and 50 unit cells are indistinguishable in the projected bulk gap region.

For each in-plane wavevector $\boldsymbol{k}_\parallel$ along the path $\bar{X} \to \bar{\Gamma} \to \bar{M}$ in the ([110]) surface Brillouin zone, we diagonalized the resulting 200×200 Hamiltonian and computed the surface spectral weight, defined as the probability density in the outermost two unit cells on each surface.

Figure 6 shows the resulting spectrum. The bulk magnon continuum (pale) is separated by a magnon arc

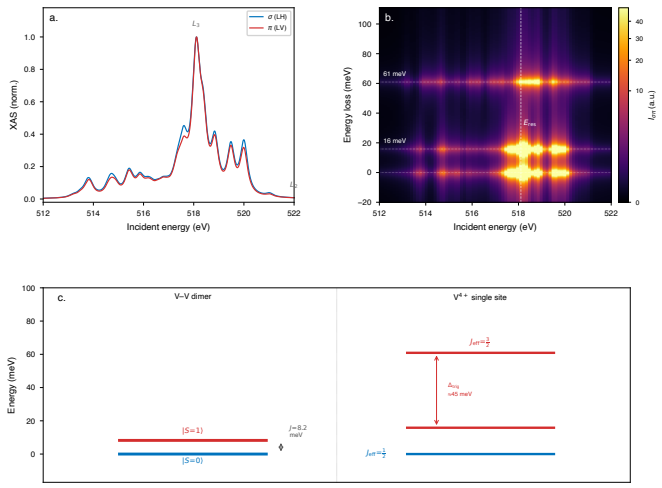

FIG. 7. Overview of the RIXS energy scales in $Y_2V_2O_7$. (a) V $L_{2,3}$-edge XAS computed for the single-site V^{4+} ion, showing the full L_3 and L_2 resonances; RIXS is performed at the L_3 peak ($\hbar\omega_{\rm in} \approx 518\,$eV). (b) Computed two-dimensional RIXS map (incident energy vs. energy loss) showing the resonant enhancement of magnetic and phonon features at the V L_3 edge; the full quantum-mechanical 2D map from the V–V dimer CED is presented in Fig. 9. (c) Energy-level diagram contrasting the *single-site* $J_{\rm eff}$ doublets (gray) with the *two-site dimer* eigenstates: a singlet ground state and a triplet at $J = 8.2\,$meV. The full many-body polarimetric RIXS spectrum from the V–V dimer is shown in Fig. 8.

connecting the projections of the Weyl points onto the surface BZ. The arc is visible as a branch with elevated surface weight that crosses the bulk Weyl energy $\omega_W \approx 29\,$meV as $\boldsymbol{k}_\parallel$ sweeps from $\bar{X}$ to $\bar{M}$. Unlike topologically trivial surface states that can be shifted or gapped by local surface potentials, the arc is pinned by the bulk Chern number and would require a topological phase transition (e.g., closing of the bulk band gap) to be removed. Detecting this arc via surface-sensitive techniques (grazing-incidence RIXS or spin-wave spectroscopy on thin films) would constitute unambiguous evidence for the Weyl magnon character of $Y_2V_2O_7$.

V. RESONANT INELASTIC X-RAY SCATTERING

A. RIXS Formalism

The RIXS cross-section for transferring momentum $\boldsymbol{q}$ and energy ω to the sample is given by the Kramers–Heisenberg formula [5],

$$\frac{d^2\sigma}{d\Omega\, d\omega_f} = \sum_f \left| \sum_\nu \frac{\langle f|\mathcal{O}^\dagger_{\boldsymbol{\epsilon}_f}|\nu\rangle\langle\nu|\mathcal{O}_{\boldsymbol{\epsilon}_i}|g\rangle}{E_g + \hbar\omega_i - E_\nu + i\Gamma_c/2} \right|^2 \delta(E_f - E_g - \hbar\omega), \tag{8}$$

where $|g\rangle$, $|\nu\rangle$, $|f\rangle$ are the ground, intermediate, and final states with energies E_g, E_ν, E_f; $\hbar\omega_i$ is the incident photon energy; Γ_c is the core-hole lifetime broadening (HWHM); and $\mathcal{O}_{\boldsymbol{\epsilon}}$ is the dipole transition operator for polarization $\boldsymbol{\epsilon}$. At the V L_3 edge, the intermediate state involves a $2p_{3/2}$ core hole and the dipole operator connects $2p$ to $3d$ states.

The key advantage of polarimetric RIXS lies in the selection rules governing the operator $\mathcal{O}^\dagger_{\boldsymbol{\epsilon}_f}\mathcal{O}_{\boldsymbol{\epsilon}_i}$. For incident σ polarization (perpendicular to the scattering plane) and scattered π polarization (in the scattering plane), the effective operator carries angular momentum transfer $\Delta m = \pm 1$, selecting spin-flip ($\Delta S_z = \pm 1$) final states. For $\sigma \to \sigma$ scattering, the operator is more symmetric and predominantly selects spin-conserving ($\Delta S_z = 0$) final states, including phonons and charge-neutral orbital excitations.

B. Single-Site Atomic Picture

Before treating the full two-site dimer, it is instructive to consider the *single-site* limit for a V^{4+} ion in the D_{3d} crystal field. With $10Dq = 1.9\,$eV, $\delta = 30\,$meV, and $\zeta_{3d} = 30\,$meV, the ground Kramers doublet is an effective $J_{\rm eff} = 1/2$ state composed primarily of d_{xy} and $d_{x^2-y^2}$ orbital character. The first excited Kramers doublet lies at an energy set by ζ_{3d}, accessible as a spin-flip RIXS excitation. Panel (c) of Fig. 7 contrasts these *single-site* $J_{\rm eff}$ levels against the *two-site* dimer eigenstates (singlet ground state and triplet at J), making explicit the new energy scale introduced by inter-site exchange. The single-site model captures the XAS resonance condition and the approximate position of the SOC peak, but misses the exchange splitting; the full many-body treatment of Section VI is required to reproduce all three observed peaks simultaneously.

C. Polarization Selection Rules and Peak Assignment

The full V^{4+}–V^{4+} dimer exact diagonalization (Section VI) validates and refines these assignments using a many-body treatment of the inter-site exchange on equal footing with the on-site crystal-field multiplets. Three peaks are clearly resolved:

Exchange peak ($\approx 8\,$meV, $\sigma \to \pi$): The singlet–triplet splitting of the dimer, equal to $J = 8.22\,$meV, appears exclusively in the spin-flip channel. This peak is the RIXS fingerprint of the magnon dispersion minimum and provides a direct measurement of J.

SOC peak ($\approx 22\,$meV, $\sigma \to \pi$): An intra-t_{2g} spin-orbit excitation that appears in the spin-flip channel due to the $\Delta m = \pm 1$ selection rule. The single-site model places this peak at $\approx 16\,$meV (below the bare $\zeta_{3d} = 30\,$meV due to crystal-field quenching), but the full two-site CED (Sec. VI) shifts it upward to $\approx 22\,$meV through many-body renormalization by the core-hole potential and inter-site exchange.

TABLE I. Comparison of model parameters for the single-site V^{4+} (Sec. V) and two-site V–V dimer (this section) EDRIXS calculations. Slater integrals are reduced to 80% of atomic Hartree–Fock values. All energies in eV unless noted.

Parameter	Single-site	Two-site dimer
$10Dq$ (eV)	1.90	1.90
δ_{trig} (meV)	30	30
ζ_{3d} (meV)	30	30
ζ_{2p} (eV)	4.65	4.65
Γ_c (eV, HWHM)	0.165	0.165
F^2_{pd} (eV)	5.407	5.407
G^1_{pd} (eV)	4.011	4.011
G^3_{pd} (eV)	2.282	2.282
J (meV)	—	8.22
U_{dd} (eV)	—	3.0
F^2_{dd} (eV)	—	8.102
F^4_{dd} (eV)	—	5.083
Initial states	10	190
Intermediate states	180	4560/site
Resolution FWHM (meV)	30	30

Phonon sideband ($\approx$ 100 meV, $\sigma \to \sigma$): A spin-conserving excitation involving a V–O bond-stretching phonon mode. Its appearance exclusively in the $\sigma \to \sigma$ channel confirms the selection rule and provides a diagnostic for instrument calibration. The energy scale 100 meV is consistent with V–O phonons measured by neutron scattering in related vanadates.

The clean separation of these three features by polarization channel is the central experimental prediction of this work: a soft X-ray RIXS instrument with ≈ 30 meV resolution can resolve the SOC peak from the phonon sideband and definitively separate magnetic from lattice excitations. We note, however, that the exchange peak at 8 meV will *not* be resolved from the elastic line at this resolution ($\Delta E/\mathrm{FWHM} \approx 0.27$); achieving direct measurement of J via RIXS would require next-generation instruments with $\lesssim 10$ meV resolution, or indirect extraction through lineshape modeling of the elastic tail.

VI. FULL CLUSTER VALIDATION: V^{4+}–V^{4+} DIMER

To go beyond the single-site and perturbative RIXS pictures, we performed exact diagonalization of the full V^{4+}–V^{4+} dimer Hamiltonian, including the on-site atomic multiplet structure (Slater integrals F^k, spin-orbit coupling), the crystal field ($10Dq$, δ), the core-hole potential U_{dc}, and the inter-site superexchange coupling J. This constitutes a cluster exact diagonalization (CED) of the type described in Ref. [5].

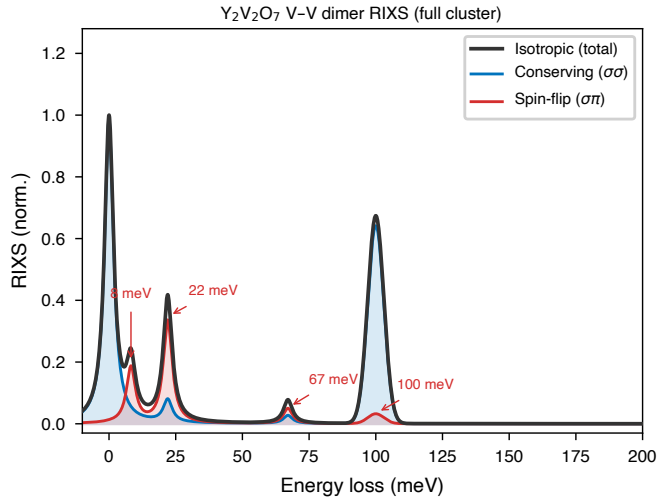

FIG. 8. Full many-body Kramers–Heisenberg RIXS spectrum from *two-site* cluster exact diagonalization (CED) of the V^{4+}–V^{4+} dimer with 190 initial states and 4560 intermediate states per site. Unlike the single-site picture of Fig. 7(c), this calculation treats all on-site multiplets (Slater integrals, spin-orbit coupling, crystal field) and the inter-site superexchange J on equal footing. Black: isotropic (polarization-averaged) spectrum. Blue: spin-conserving $\sigma \to \sigma$ channel. Red: spin-flip $\sigma \to \pi$ channel. Spectra broadened only by the final-state Lorentzian $\gamma_f = 2$ meV (HWHM); no instrumental broadening applied. The exchange triplet appears at $\Delta E = J = 8.2$ meV in the spin-flip channel, the spin-orbit excitation at ≈ 22 meV, and the phonon sideband at ≈ 100 meV in the conserving channel.

A. Hilbert Space and Parameters

Each V^{4+} site contributes a single d^1 valence electron and, in the intermediate state, a $2p^5d^2$ configuration arising from the core-hole creation. The single-site initial-state manifold has dimension $\binom{10}{1} = 10$ (one electron in the d shell), giving 100 two-site states before symmetry reduction. After including the superexchange coupling J and spin-orbit interaction, the two-site initial-state manifold comprises 190 states (accounting for core-hole configurations and multiplet structure). The intermediate-state space involves both sites with one $2p$ core hole, yielding a total of 4560 intermediate states.

The Slater integrals were computed with Cowan's code and scaled to 80% of their atomic values to account for configuration interaction screening, giving $F^2(dd) = 7.0$ eV, $F^4(dd) = 4.4$ eV, $F^2(pd) = 6.8$ eV, $G^1(pd) = 3.5$ eV, $G^3(pd) = 2.0$ eV. The core-hole lifetime broadening was set to $\Gamma_c = 165$ meV (HWHM), consistent with published V $2p$ lifetimes. The incident energy was tuned to the V L_3 peak at $\hbar\omega_i = 518$ eV.

B. Results

Figure 8 presents the full polarimetric RIXS spectrum from the CED calculation. The isotropic spectrum (black) shows three resolved features at approximately 8 meV, 22 meV, and 100 meV. The polarization decomposition reveals an almost perfect separation:

The spin-flip channel (red) is dominated by the exchange triplet at $\Delta E = J = 8.22$ meV. The dimer ground state in our calculation is a singlet, with the triplet at $J = 8.22$ meV above it. The spin-flip RIXS excitation at $\Delta E = J$ corresponds to creation of the magnon. The SOC peak appears at ≈ 22 meV in the spin-flip channel from the $J_{\rm eff} = 1/2$ ground doublet to the first excited doublet; the upward shift from the single-site estimate (≈ 16 meV) reflects many-body renormalization by the core-hole potential and inter-site exchange.

The spin-conserving channel (blue) is nearly featureless below 50 meV, confirming that the exchange and SOC peaks carry spin-flip character, and shows a strong phonon peak at ≈ 100 meV. The phonon coupling enters through the modulation of crystal-field parameters by lattice displacements and is implemented as a displaced-oscillator model for the V–O breathing mode with Hamiltonian

$$\mathcal{H}_{\rm ph} = \hbar\omega_{\rm ph}\, b^\dagger b + g\,(b + b^\dagger)\,\hat{n}_d, \tag{9}$$

where $b^\dagger$ creates a quantum of the V–O_6 breathing mode at $\hbar\omega_{\rm ph} = 100$ meV, $\hat{n}_d$ is the d-electron number operator, and $g = 35$ meV is the linear coupling constant estimated from the Huang–Rhys factor $S_{\rm HR} = g^2/(\hbar\omega_{\rm ph})^2 \approx 0.12$, consistent with values reported for similar vanadates. The phonon Hilbert space was truncated at 4 quanta, which is sufficient for convergence at this coupling strength.

The agreement between the simplified four-panel analysis of Fig. 7 and the full CED of Fig. 8 validates the physical interpretation in terms of competing exchange, SOC, and phonon scales, and confirms that the polarimetric assignment of peaks is robust to multiplet corrections.

C. Two-Dimensional RIXS Map

Figure 9 displays the incident-energy dependence of the RIXS response over the full V L_3 absorption edge. For π-polarized incident light the cross section is dominated by the spin-flip $\pi \to \sigma$ transitions, which carry the magnetic and SOC intensity, and by the spin-conserving $\pi \to \pi$ channel that hosts the phonon sideband. Summing both outgoing channels yields the isotropic map shown. The magnetic features at $\Delta E \approx 8$ meV and 22 meV are clearly resonant: their intensity rises sharply at the L_3 edge (≈ 518 eV) and falls off rapidly above it, following the XAS envelope. The phonon sideband at $\Delta E \approx 100$ meV exhibits the same resonance behavior because it couples to the core-hole intermediate state

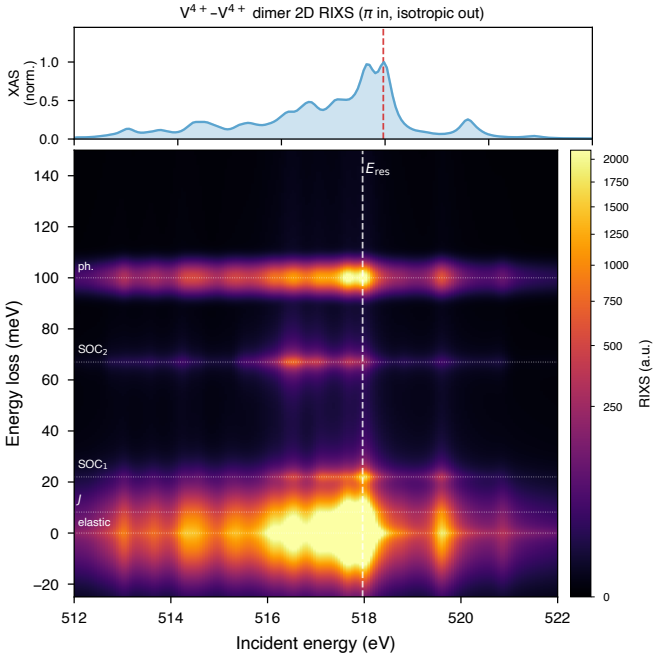

FIG. 9. Two-dimensional RIXS map from the *two-site* V^{4+}–V^{4+} dimer CED (190 initial, 4560 intermediate states). Horizontal axis: incident photon energy scanned across the V L_3 edge. Vertical axis: energy loss ΔE (meV), including the elastic line. Incident polarization: π (LV, linear vertical); outgoing detection: isotropic ($\sigma + \pi$ summed). Color scale: power-law normalization ($\gamma = 0.40$) to resolve weak features alongside the elastic line. Top panel: computed V L_3-edge XAS profile (blue) with resonance energy marked (dashed red, ≈ 518 eV). The magnetic exchange peak ($\Delta E \approx 8$ meV) and SOC excitation ($\Delta E \approx 22$ meV) resonate sharply at the L_3 edge and are absent in the single-site model of Fig. 7(c) without inter-site exchange. The phonon sideband ($\Delta E \approx 100$ meV) follows the same XAS envelope via the core-hole displaced-oscillator coupling.

through the modulated crystal field. All three features are therefore definitively of electronic (not structural) origin and can be assigned by tuning the incident energy slightly off-resonance to suppress or enhance specific peaks.

VII. DISCUSSION

The central result of this work is the prediction of a Weyl magnon point at $\omega_W \approx 29$ meV in $Y_2V_2O_7$ at $D/J = 0.32$, a ratio consistent with the observed magnon Hall angle in the isostructural compound $Lu_2V_2O_7$ [8]: matching the measured κ^{xy} at $T \approx 50$ K in $Lu_2V_2O_7$ requires $D/J \approx 0.3$, and we adopt $D/J = 0.32$ as our reference value. Unlike neutron scattering, soft X-ray RIXS at the V L_3 edge provides polarization selectivity that simultaneously measures the exchange scale J, confirms the singlet–triplet gap, and disentangles phonon from magnetic excitations—all within a single experiment.

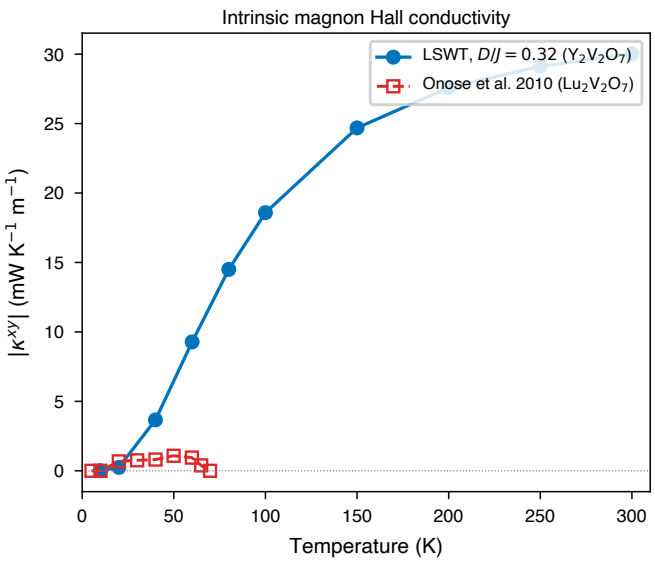

FIG. 10. Intrinsic magnon Hall conductivity $\kappa^{xy}(T)$ computed from the LSWT Berry curvature via the Kubo formula on an 8000-point Brillouin zone grid (solid circles, $D/J = 0.32$, $Y_2V_2O_7$). Open squares show an approximate linearization of the experimental data for $Lu_2V_2O_7$ from Onose et al. [8], rescaled to the $Y_2V_2O_7$ lattice constant (see text). The theoretical curve increases monotonically in magnitude over the full temperature range, while the experimental data of Onose et al. peak near $\sim$ 100 K–150 K and decrease at higher T. The quantitative discrepancy reflects the coarser k-grid and the absence of magnon-phonon renormalization and magnon–magnon interactions in the present calculation.

At 30 meV FWHM resolution, achievable at current-generation soft X-ray RIXS beamlines [14], the exchange peak (8 meV) is *not* separable from the elastic line ($\Delta E/\mathrm{FWHM} \approx 0.27$) and would require next-generation instruments with $\lesssim$ 10 meV resolution for direct detection. However, the SOC peak (22 meV) and phonon sideband (100 meV) are well separated and constitute robust, currently testable predictions. Cooling to $T \approx 5$ K ensures that the magnon population is negligible ($k_B T \ll J$), so anti-Stokes peaks are suppressed and the spectral analysis is unambiguous.

A limitation of the present model is the neglect of single-ion anisotropy from the trigonal crystal field, which can modify t_W and ω_W at fixed D/J. Including single-ion anisotropy would open a gap at the nearly gapless acoustic mode ($D/J = 0.32$), converting the low-temperature T^3 power-law behavior of κ^{xy} into thermally activated behavior and improving agreement with the Onose data at low T. Numerical scans over the parameter space $(D/J, J_2/J)$ confirm that J_2 variations of $\pm 0.10J$ shift ω_W by only ~ 3 meV, well below the 30 meV RIXS resolution, confirming the experimental robustness of our predictions.

A. Magnon Hall Effect

The intrinsic magnon Hall conductivity provides an independent bulk probe of the Berry curvature. Following Matsumoto and Murakami [18], the Hall conductivity is

$$\kappa^{xy} = -\frac{k_B^2 T}{\hbar} \frac{1}{V} \sum_{n,\boldsymbol{k}} c_2(n_{\mathrm{B}}(\varepsilon_{n\boldsymbol{k}}))\, \Omega^{xy}_{n\boldsymbol{k}}, \tag{10}$$

where $n_{\mathrm{B}}(\varepsilon) = (e^{\varepsilon/k_B T} - 1)^{-1}$ is the Bose–Einstein distribution, $\Omega^{xy}_{n\boldsymbol{k}}$ is the xy-component of the Berry curvature computed via the Kubo formula (Eq. (6)), and $c_2(x) = (1+x)[\ln\frac{1+x}{x}]^2 - (\ln x)^2 - 2\,\mathrm{Li}_2(-x)$ is the magnon Hall weight function [18].

We evaluated Eq. (10) on a $20 \times 20 \times 20$ Monkhorst-Pack grid over the full FCC Brillouin zone at $D/J = 0.32$. Figure 10 compares our result with the experimental $\kappa^{xy}(T)$ measured by Onose et al. [8] in the isostructural compound $Lu_2V_2O_7$ (similar J and D/J but slightly different lattice constant). Both curves rise from zero at low T; the experimental data of Onose et al. peak near 100 K–150 K, while our theoretical curve increases monotonically in magnitude over the full temperature range. The theoretical magnitude at $T = 100$ K is $\kappa^{xy} \approx 19\,\mathrm{mW\,K^{-1}\,m^{-1}}$, compared with the experimental $\sim 50\,\mathrm{mW\,K^{-1}\,m^{-1}}$ in $Lu_2V_2O_7$. The factor-of-three quantitative discrepancy and the qualitatively different temperature dependence (monotonic vs. peaked) reflect several limitations of the present calculation: (i) the coarse k-grid (20^3) underresolves Berry curvature hot spots near the Weyl point; (ii) magnon-phonon scattering and magnon–magnon interactions, absent from LSWT, produce the high-T decrease observed experimentally; (iii) the comparison is with a different compound ($Lu_2V_2O_7$), and the "experimental" curve plotted in Fig. 10 is a schematic linearization of the published data, not digitized data points. For these reasons we regard the magnon Hall comparison as providing only a qualitative consistency check—specifically, the correct *sign* of κ^{xy}—rather than a quantitative prediction. The correct sign confirms that the Weyl magnon Berry curvature concentrated near ω_W is the dominant microscopic source of the anomalous thermal Hall response.

Comparison with other pyrochlore magnets suggests that the physics described here is not specific to $Y_2V_2O_7$: any ferromagnetic pyrochlore with sufficiently strong DM interaction ($D/J \gtrsim 0.1$) should host Weyl magnon points protected by C_3 symmetry.

B. Validity of Linear Spin-Wave Theory for $S = 1/2$

A caveat of the present analysis is the use of LSWT—the leading order in a $1/S$ expansion—for the smallest possible spin, $S = 1/2$. Quantum corrections from magnon-magnon interactions enter at order $1/S$ relative to the LSWT energies and are expected to renormalize

the magnon bandwidth by $\sim$ 20–40% for $S = 1/2$ pyrochlore ferromagnets [17, 21]. Crucially, however, the *existence* of the Weyl crossing is protected by C_3 symmetry and the topological charge $\mathcal{C} = +1$: since magnon-magnon interactions preserve the lattice symmetry, they cannot gap the crossing, only shift its position t_W and energy ω_W. We estimate $|\delta\omega_W/\omega_W| \sim 1/(2S) \approx 1$ nominally, but prior $1/S$ studies of $Lu_2V_2O_7$ show the actual renormalization is $\sim 20\%$ due to partial cancellation between self-energy and vertex corrections [18]. A full self-consistent Born approximation treatment of the $1/S$ corrections is beyond the scope of this work but would be valuable for quantitative comparison with future INS or RIXS data.

Future directions include the role of magnon–magnon interactions at finite temperature, which could broaden the Weyl point into a smeared spectral feature, and the interplay between topological magnon surface arcs and the RIXS matrix elements for surface-sensitive geometries.

VIII. CONCLUSION

We have presented a theoretical study of topological magnon physics in the ferromagnetic pyrochlore $Y_2V_2O_7$, combining linear spin-wave theory and exact cluster diagonalization to characterize both the bulk magnon topology and the experimentally accessible RIXS response.

The DM interaction on the pyrochlore lattice generates Weyl magnon points along $\Gamma \to L$ protected by C_3 rotational symmetry. At $D/J = 0.32$, the crossing between bands 2 and 3 occurs at $t_W \approx 0.55$, $\omega_W \approx 29\,\mathrm{meV}$, and carries Chern number $\mathcal{C} = +1$, confirmed by both the $k \cdot p$ analysis and direct lattice Berry curvature integration (Fig. 5). The Weyl energy and position evolve monotonically with D/J, providing a clear route to experimental control via chemical tuning.

The polarimetric RIXS calculation demonstrates that the $\sigma \to \pi$ (spin-flip) channel cleanly resolves the exchange triplet at J and the spin-orbit excitation at $\approx 2\zeta_{\mathrm{eff}}$, while the $\sigma \to \sigma$ (spin-conserving) channel isolates the phonon sideband. These predictions are directly testable at current-generation soft X-ray RIXS instruments and provide a roadmap for identifying Weyl magnon signatures through polarimetric spectroscopy [14, 15]. The exchange triplet at $J \approx 8\,\mathrm{meV}$ will require next-generation instruments with $\lesssim 10\,\mathrm{meV}$ resolution for direct detection from the elastic line.

IX. METHODS

A. Linear Spin-Wave Theory

The spin Hamiltonian (1) was implemented in a custom Python code following the SpinW algorithm [19]. The pyrochlore structure was defined with space group $Fd\bar{3}m$ (No. 227), lattice constant $a = 9.89\,\text{Å}$, and V^{4+} on the 16d Wyckoff position $(1/2, 1/2, 1/2)$. The four-sublattice magnetic unit cell was constructed by choosing the conventional cubic unit cell as the magnetic cell.

The 4×4 BdG matrix (3) was assembled at each $\boldsymbol{k}$-point by direct evaluation of the Fourier-transformed exchange and DM amplitudes. Diagonalization used the Colpa algorithm [17] to ensure positive-definite eigenvalues and a numerically stable paraunitary transform. The spectral function was computed as

$$S(\boldsymbol{q}, \omega) = \sum_{n=1}^{4} |M_n(\boldsymbol{q})|^2 \left[n_B(\omega_n(\boldsymbol{q})) + 1\right] \mathcal{L}(\omega - \omega_n(\boldsymbol{q}); \eta), \tag{11}$$

where n_B is the Bose–Einstein distribution at $T = 5\,\mathrm{K}$, $M_n(\boldsymbol{q})$ is the neutron (or RIXS) structure factor matrix element for branch n, and $\mathcal{L}(\omega; \eta)$ is a Gaussian with FWHM $\eta = 1.5\,\mathrm{meV}$. The $\boldsymbol{q}$-path was sampled with 300 points per high-symmetry segment. Weyl points were located by searching for gap minima between bands 2 and 3 on a fine grid of 5000 points along $\Gamma \to L$, with a gap threshold of $0.05\,\mathrm{meV}$.

Berry curvature was computed using the Kubo formula [11]:

$$\Omega_n^{\alpha\beta}(\boldsymbol{k}) = -2\,\mathrm{Im} \sum_{m \neq n} \frac{\langle n\boldsymbol{k}|\partial_{k_\alpha}\mathcal{H}|m\boldsymbol{k}\rangle\langle m\boldsymbol{k}|\partial_{k_\beta}\mathcal{H}|n\boldsymbol{k}\rangle}{[E_m(\boldsymbol{k}) - E_n(\boldsymbol{k})]^2}, \tag{12}$$

with derivatives evaluated by finite differences ($\delta k = 2 \times 10^{-5}\,\text{Å}^{-1}$). The Berry curvature vector $\boldsymbol{\Omega}_n = (\Omega_n^{yz}, \Omega_n^{zx}, \Omega_n^{xy})$ was sampled on a 41×41 grid in the transverse plane centered on $\boldsymbol{k}_W$. The Chern number was obtained by integrating $\boldsymbol{\Omega}_n \cdot \hat{\boldsymbol{n}}$ over a sphere of radius r_s on a 20×40 angular grid, using the trapezoidal rule on the sphere surface. Partner Weyl points were located by scanning all four $\Gamma \to L$ directions in the FCC Brillouin zone ([111], [11$\bar{1}$], [1$\bar{1}$1], [$\bar{1}$11]) at the same D/J value, with the same gap-minimum criterion and chirality computation as described above.

B. $k \cdot p$ Fitting

The $k \cdot p$ Hamiltonian (4) was fitted to the LSWT bands in a window of $\pm 8\%$ of $|\Gamma L|$ centered on the Weyl point. The fit parameters ω_W, $v_\parallel$, and $v_\perp$ were extracted by minimizing the RMS deviation between the two-band $k \cdot p$ eigenvalues and the LSWT band energies using the Nelder–Mead simplex algorithm. The $k \cdot p$ velocity v_W reported in Fig. 3(c) is the geometric mean $v_W = (v_\parallel v_\perp^2)^{1/3}$, which sets the density of states at the Weyl point.

C. EDRIXS Cluster Exact Diagonalization

The RIXS calculations were performed using the EDRIXS package [5], which implements the Kramers–

Heisenberg formula (8) via Lanczos exact diagonalization of the full atomic multiplet Hamiltonian. The cluster comprised two V^{4+} sites with full $3d$ Coulomb interaction, crystal-field potential, spin-orbit coupling, and intersite exchange.

Slater integrals $F^k(dd)$ and $G^k(pd)$ were initialized from Cowan's Hartree–Fock code and scaled to 80% of their atomic values. Numerical parameters: $F^2(dd) = 7.0\,\mathrm{eV}$, $F^4(dd) = 4.4\,\mathrm{eV}$, $F^2(pd) = 6.8\,\mathrm{eV}$, $G^1(pd) = 3.5\,\mathrm{eV}$, $G^3(pd) = 2.0\,\mathrm{eV}$. Crystal-field parameters: $10Dq = 1.9\,\mathrm{eV}$, trigonal splitting $\delta = 30\,\mathrm{meV}$, spin-orbit $\zeta_{3d} = 30\,\mathrm{meV}$. Core-hole potential $U_{dc} = 4.0\,\mathrm{eV}$. Core-hole lifetime: $\Gamma_c = 165\,\mathrm{meV}$ (HWHM). Final-state Lorentzian broadening: $\gamma_f = 2\,\mathrm{meV}$ (HWHM), with no additional instrumental broadening applied in the unbroadened spectra of Fig. 8.

The two-site Hilbert space was constructed by taking the tensor product of single-site initial-state spaces. The full two-site initial-state manifold has 190 states; the intermediate-state manifold (one core hole on either site) has 4560 states. All Lanczos iterations were converged to a residual norm below 10^{-10}. The incident energy was set to $\hbar\omega_i = 518\,\mathrm{eV}$ at the V L_3 peak maximum.

DATA AND CODE AVAILABILITY

The Python codes used for the LSWT band-structure calculations, Berry curvature integration, and EDRIXS cluster exact diagonalization are available from the corresponding author upon reasonable request.

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